

Euro Area Recession Risk Indicator

Replicating Giannone & Furno (Nowcasting Recession Risk)

Automated Report

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Welcome to the **Euro Area Recession Risk Tracker**. This site presents real-time nowcasts of recession probabilities for the Euro Area (EA) and its four largest member states: Germany (DE), France (FR), Italy (IT), and Spain (ES).

Our methodology builds on the Bayesian logit nowcasting framework proposed by Furno and Giannone (2024). By utilizing a **Hierarchical Bayesian model**, we allow the model to share information across different regions. This hierarchical structure (“shrinkage”) is particularly valuable for improving the reliability of country-specific nowcasts during periods of high economic uncertainty.

```
# Install packages if missing
# install.packages(c("eurostat", "ecb", "dplyr", "tidyr", "lubridate",
#                   "ggplot2", "patchwork", "brms", "zoo"))

library(eurostat)
library(ecb)
library(dplyr)
library(tidyr)
library(lubridate)
library(ggplot2)
library(patchwork)
library(brms)
library(zoo)
```

```

# Target countries: Euro Area (EA), Germany (DE), France (FR), Italy (IT), Spain (ES)
# Note: Eurostat uses 'EA' or 'EA19'/'EA20', ECB uses 'U2' for the Euro Area.
target_geo_eurostat <- c("EA20", "EA19", "DE", "FR", "IT", "ES")
target_geo_ecb <- c("U2", "DE", "FR", "IT", "ES")

## 1. Download GDP Data (Eurostat)

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# 'namq_10_gdp' is the Eurostat dataset for GDP and main components
gdp_raw <- get_eurostat("namq_10_gdp", time_format = "date", stringsAsFactors = FALSE)

gdp <- gdp_raw %>%
  # Safely rename new API columns to the old names, ignoring them if they don't exist
  rename(any_of(c(time = "TIME_PERIOD", values = "OBS_VALUE"))) %>%
  filter(
    na_item == "B1GQ",          # Gross domestic product at market prices
    s_adj == "SCA",             # Seasonally and calendar adjusted data
    unit == "CLV_PCH_PRE",      # Percentage change on previous period (QoQ)
    geo %in% target_geo_eurostat
  ) %>%
  select(gdp_growth = values) %>%
  mutate(
    # Standardize Euro Area code to 'EA'
    geo = ifelse(grepl("EA", geo), "EA", geo),
    quarter = as.yearqtr(time)
  ) %>%
  # In case of overlapping EA definitions, keep the longest non-NA series
  group_by(geo, quarter) %>%
  summarise(gdp_growth = mean(gdp_growth, na.rm = TRUE), .groups = "drop")

# Download CISS Data
# The CISS key structure: CISS.Frequency.ReferenceArea...
# Using wildcard for the country/area to get all available
ciss_key <- "CISS.D..ZOZ.4F.EC.SS_CIN.IDX"
ciss_raw <- get_data(ciss_key)

ciss <- ciss_raw %>%
  filter(ref_area %in% target_geo_ecb) %>%
  mutate(
    obstime_char = as.character(obstime),
    time = as.Date(ifelse(nchar(obstime_char) == 7, paste0(obstime_char, "-01"), obstime_char

```

```

    geo = ifelse(ref_area == "U2", "EA", ref_area),
    quarter = as.yearqtr(time)
  ) %>%
  group_by(geo, quarter) %>%
  summarise(ciss = mean(obsvalue, na.rm = TRUE), .groups = "drop")

## 3. Download Economic Sentiment Indicator (Eurostat)

esi_raw <- get_eurostat("ei_bssi_m_r2", time_format = "date", stringsAsFactors = FALSE)

esi <- esi_raw %>%
  # Safely rename new API columns to the old names, ignoring them if they don't exist
  rename(any_of(c(time = "TIME_PERIOD", values = "OBS_VALUE"))) %>%
  filter(
    indic == "BS-ESI-I",      # Economic Sentiment Indicator
    s_adj == "SA",           # Seasonally adjusted
    geo %in% target_geo_eurostat
  ) %>%
  select(geo, time, esi = values) %>%
  mutate(
    geo = ifelse(grepl("EA", geo), "EA", geo),
    quarter = as.yearqtr(time)
  ) %>%
  group_by(geo, quarter) %>%
  summarise(es_i = mean(es_i, na.rm = TRUE), .groups = "drop")

# Combine data and calculate recession risk
# Merge all datasets
df <- gdp %>%
  inner_join(ciss, by = c("geo", "quarter")) %>%
  inner_join(es_i, by = c("geo", "quarter")) %>%
  arrange(geo, quarter) %>%
  group_by(geo) %>%
  mutate(
    # Recession definition: current quarter < 0 AND previous quarter < 0
    recession = ifelse(gdp_growth < 0 & lag(gdp_growth, 1) < 0, 1, 0)
  ) %>%
  drop_na() %>% # Drop NAs created by lag and missing series overlaps
  ungroup()

# Standardize the predictors to help the Bayesian model converge faster
df <- df %>%

```

```

mutate(
  ciss_scaled = scale(ciss)[,1],
  esi_scaled = scale(esi)[,1]
)

# Hierarchical Bayesian Logit model
# Determine the cut-off date (5 years ago from the maximum available date in the dataset)
max_date <- max(df$quarter)
cutoff_date <- max_date - 5 # 5 years subtraction on yearqtr object

# Split data
train_data <- df %>% filter(quarter <= cutoff_date)
inference_data <- df # We infer on the whole set

# Hierarchical Bayesian Logit Model using brms
# Random intercepts and random slopes for each geo (Euro Area and countries)
formula <- brmsformula(
  recession ~ ciss_scaled + esi_scaled + (1 + ciss_scaled + esi_scaled | geo),
  family = bernoulli(link = "logit")
)

# Fit the model (this may take a minute or two depending on your machine)
# We set a seed for reproducibility.
brms_model <- brm(
  formula = formula,
  data = train_data,
  prior = c(
    prior(normal(0, 5), class = "Intercept"),
    prior(normal(0, 2), class = "b")
  ),
  chains = 2,
  cores = 2,
  iter = 2000,
  seed = 123,
  silent = 2,
  refresh = 0
)

summary(brms_model)

```

Family: bernoulli
 Links: mu = logit

Formula: recession ~ ciiss_scaled + esi_scaled + (1 + ciiss_scaled + esi_scaled | geo)
 Data: train_data (Number of observations: 572)
 Draws: 2 chains, each with iter = 2000; warmup = 1000; thin = 1;
 total post-warmup draws = 2000

Multilevel Hyperparameters:
 ~geo (Number of levels: 5)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS
sd(Intercept)	1.04	0.67	0.08	2.75	1.00	433
sd(ciiss_scaled)	0.37	0.34	0.01	1.26	1.01	355
sd(es_i_scaled)	0.54	0.58	0.02	2.70	1.00	420
cor(Intercept,ciiss_scaled)	0.04	0.49	-0.85	0.88	1.01	1863
cor(Intercept,esi_scaled)	-0.13	0.49	-0.91	0.82	1.00	1710
cor(ciiss_scaled,esi_scaled)	0.08	0.51	-0.86	0.91	1.00	1271

	Tail_ESS
sd(Intercept)	387
sd(ciiss_scaled)	1079
sd(es_i_scaled)	164
cor(Intercept,ciiss_scaled)	1201
cor(Intercept,esi_scaled)	1159
cor(ciiss_scaled,esi_scaled)	1426

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-3.39	0.60	-4.75	-2.12	1.00	618	779
ciiss_scaled	0.64	0.27	0.11	1.26	1.00	792	387
esi_scaled	-1.41	0.34	-2.05	-0.71	1.00	637	300

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
# Out of sample inference
# Extract predicted probabilities (Estimate = mean of posterior)
predictions <- fitted(brms_model, newdata = inference_data, type = "response")

df_results <- inference_data %>%
  mutate(
    probb_recession = predictions[, "Estimate"],
    lower_95 = predictions[, "Q2.5"],
    upper_95 = predictions[, "Q97.5"],
    is_forecast = ifelse(quarter > cutoff_date, "Inferred (Last 5 Yrs)", "Training Data")
  )
```

)

1 Dashboard Overview

This interactive dashboard displays the estimated recession probabilities derived from our Hierarchical Bayesian Logit model. By combining macroeconomic sentiment (ESI) and financial conditions (CISS) within a unified hierarchical structure, the model estimates the probability that the Euro Area and its four largest economies are in a technical recession.

1.1 How to Read the Charts

- **Recession Probability (Solid Line):** Represents the posterior mean probability of being in a recession.
- **95% Credible Intervals (Shaded Ribbons):** Reflect the uncertainty in our estimates. The wider the ribbon, the more uncertainty there is in the estimate.
- **Historical Recessions (Grey Shaded Bars):** Indicate periods of actual recessions defined by the technical definition of two consecutive quarters of negative quarter-on-quarter GDP growth.
- **Data Splitting (Red Dashed Line):** Denotes the out-of-sample cutoff.
 - **Blue Line/Ribbon (Training Data):** In-sample model performance.
 - **Orange Line/Ribbon (Inferred Data):** Out-of-sample estimates for the last 5 years, showing how the model performs on unseen data.

2 Model Estimation Results

Below, we run the visualization routine to plot the estimated probabilities. The top panel displays the results for the aggregate Euro Area, while the bottom panel shows the individual country-level results for Germany, France, Italy, and Spain.

```
# Data visualization
# Function to generate a plot for a specific geography
plot_geo <- function(data, region_name, is_main = FALSE) {

  p_data <- data %>% filter(geo == region_name)

  p <- ggplot(p_data, aes(x = as.Date(quarter))) +
    # Actual Recession shaded areas
    geom_rect(
```

```

    data = p_data %>% filter(recession == 1),
    aes(xmin = as.Date(quarter) - 45, xmax = as.Date(quarter) + 45,
        ymin = 0, ymax = 1),
    fill = "grey80", alpha = 0.5, inherit.aes = FALSE
  ) +
  # Recession Probability line
  geom_line(aes(y = prob_recession, color = is_forecast), size = 1) +
  # 95% Credible Intervals
  geom_ribbon(aes(ymin = lower_95, ymax = upper_95, fill = is_forecast), alpha = 0.2) +
  geom_vline(xintercept = as.Date(cutoff_date), linetype = "dashed", color = "red") +
  scale_y_continuous(labels = scales::percent_format(), limits = c(0, 1)) +
  scale_color_manual(values = c("Training Data" = "#004B87", "Inferred (Last 5 Yrs)" = "#E377C2")) +
  scale_fill_manual(values = c("Training Data" = "#004B87", "Inferred (Last 5 Yrs)" = "#E377C2")) +
  labs(
    title = region_name,
    y = "Recession Probability",
    x = "",
    color = "Data Period",
    fill = "Data Period"
  ) +
  theme_minimal() +
  theme(
    legend.position = "bottom",
    plot.title = element_text(face = "bold", size = ifelse(is_main, 16, 12))
  )

  return(p)
}

# 1. Main plot for Euro Area
p_ea <- plot_geo(df_results, "EA", is_main = TRUE) +
  labs(subtitle = "Grey bars indicate actual historical recessions (2 consecutive quarters of decline)")

# 2. Country plots
p_de <- plot_geo(df_results, "DE")
p_fr <- plot_geo(df_results, "FR")
p_it <- plot_geo(df_results, "IT")
p_es <- plot_geo(df_results, "ES")

# Combine country plots into a 2x2 grid
country_grid <- (p_de | p_fr) / (p_it | p_es) +
  plot_layout(guides = "collect") &

```

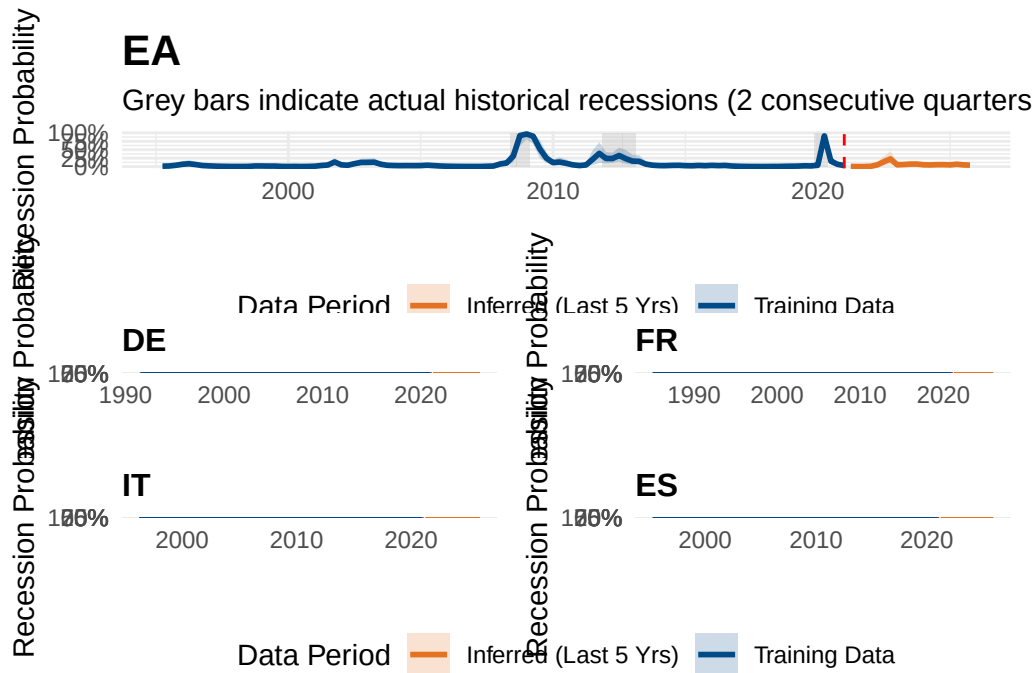
```

theme(legend.position = 'bottom')

# Final layout: EA on top, 2x2 grid below
final_plot <- p_ea / country_grid + plot_layout(heights = c(1, 1.5))

final_plot

```



Furno, Francesco, and Domenico Giannone. 2024. "Nowcasting Recession Risk." *Research Methods and Applications on Macroeconomic Forecasting*. <https://www.amazon.science/publications/nowcasting-recession-risk>.